



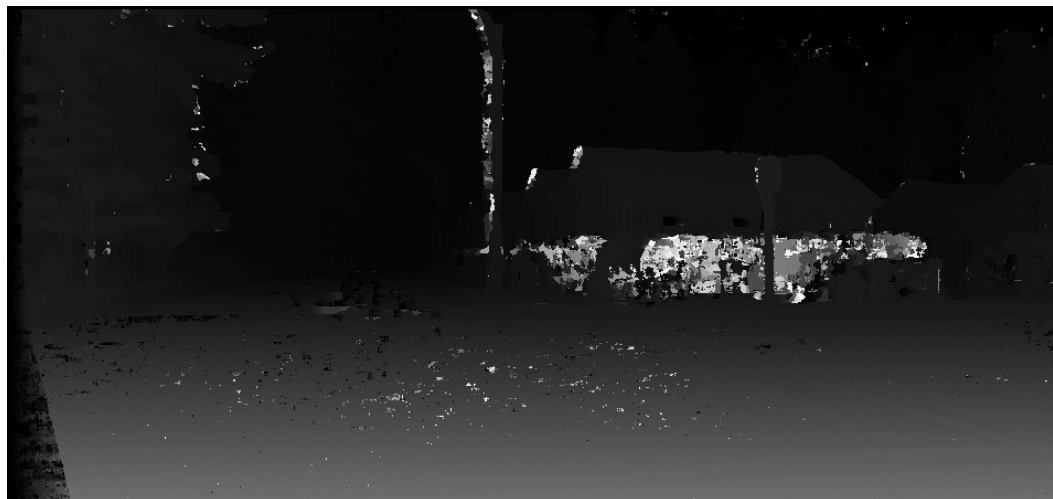
UNIVERSITÀ DI PARMA

Laboratory Stereo Matching

- We have two stereo and rectified images
 - L.pgm and R.pgm



- Develop the following function
 - `void mySAD_Disparity7x7(const cv::Mat &l, const cv::Mat &r, cv::Mat &out)`



- Use a 7x7 neighborhood on the left image to match against right candidates
- Where the SAD match reaches the minimum value take it as the best correspondence
- Save disparity d as CV_32FC1
- Limit the search range as $[0, 127]$
- Hints
 - We can solve this with 5 nested cycles

- Starting from the computed disparity create a new `cv::Mat vdisp` using
 - `void VDisparity(const cv::Mat &disp, cv::Mat &out);`
 - Height $\rightarrow$ the same as disparity
 - Width $\rightarrow$ 128
- Each row represents an histogram of the values of the same row in the disparity image
- Namely each pixel encodes how many matches we have at a given disparity for that row

